

Approach

The prototype is designed as a lightweight label-verification assistant that compares information from an alcohol label image against application data provided by the user.

The workflow is:

1. The user enters application details such as brand name, class/type, alcohol content, and net contents.
2. The user uploads a label image.
3. The system performs image preprocessing and OCR to extract text from the label.
4. Extracted values are normalized and compared with the expected application values.
5. Compliance rules are applied to fields that require stricter validation, such as the Government Health Warning.
6. The application returns one of three outcomes: **Pass, Review, or Fail**, along with the extracted value, expected value, and reason for the result.

The design intentionally separates OCR from compliance logic. OCR is used to read the label, while deterministic rules perform most of the actual verification. This reduces dependence on generative AI and makes results easier for compliance agents to understand and audit.

The prototype also supports batch-oriented processing so that multiple label applications can be reviewed more efficiently.

Tools Used

Frontend

- Next.js / React
- TypeScript
- Tailwind CSS

Backend

- Python
- FastAPI
- Pydantic

Image Processing

- OpenCV
- Pillow

OCR

- RapidOCR / ONNX Runtime

Validation

- Python-based normalization and rule engine
- Regular expressions for extracting ABV, proof, volume, and warning text
- Fuzzy/string similarity only where minor formatting differences are acceptable

Development and Deployment

- GitHub for source control
- Docker for reproducible deployment
- Vercel for the frontend
- Render or another container-compatible service for the FastAPI/OCR backend

Key Design Decisions

The prototype was designed around the stakeholder requirements identified in the discovery interviews.

Performance: The target is to return results within approximately five seconds for a normal single-label request. OCR models are loaded once when the backend starts, and the critical verification path does not require calls to an external LLM.

Human-in-the-loop review: The system does not automatically reject every uncertain result. Low OCR confidence, poor image quality, or ambiguous matches are classified as **Review**, allowing an agent to make the final determination.

Flexible matching: Fields such as brand name are normalized for capitalization, whitespace, and minor formatting differences. For example, `Stone's Throw` and `STONE'S THROW` should normally be treated as equivalent.

Strict regulatory checks: Fields requiring exact wording, such as the Government Health Warning, use deterministic validation rather than fuzzy matching.

Explainability: Every result shows the expected value, detected value, result, and reason so that compliance agents can quickly understand why a field passed or failed.

Limited external dependencies: OCR and validation can run locally within the application environment. This was chosen to reduce latency and avoid reliance on external AI endpoints that could be blocked in a restricted government network.

Assumptions

- This project is a proof of concept and is not intended to make final legal or regulatory compliance decisions.
- Final approval or rejection remains the responsibility of an authorized compliance reviewer.
- The initial prototype focuses on a limited set of common fields: brand name, class/type, alcohol content, net contents, and Government Health Warning.

- Users provide the expected application data directly or through a structured batch file such as CSV.
- Uploaded images are assumed to be reasonably readable photographs or scans of alcohol labels.
- The application does not integrate directly with the existing COLA system.
- Files are processed temporarily and are not intended to be permanently stored by the prototype.
- Image-quality problems such as blur, glare, poor lighting, or extreme perspective may reduce OCR confidence and cause the result to be flagged for manual review.
- Formatting checks such as bold text are treated as confidence-based visual checks rather than guaranteed typographic verification.
- Additional beverage-specific TTB rules could be added later as separate validation modules for beer, wine, and distilled spirits.

Future Improvements

Potential next steps include direct COLA integration, more advanced label-region detection, improved perspective and glare correction, beverage-specific rule sets, configurable compliance policies, role-based access control, audit logging, and production deployment within an authorized government cloud environment.